


Rating
Buy

Japan
Japan

Machinery

Company
Harmonic Drive Systems

Reuters
6324.T

Bloomberg
6324 JP

Exchange
TYO

Ticker
6324

Date
8 August 2026

Forecast Change

Price at 6 Aug 2026 (JPY)	7,130
Price target - 12mth (JPY)	9,900
52-week range (JPY)	8,890 - 2,328

Beat and raise in FY1Q; Semis and medical strong; AI robot growing; BUY

Harmonic Drive Systems (HDS) reported 1Q FY3/27 (3M ending June 2026) results after market close on 7 August.

In FY1Q, HDS reported profits ahead of expectations on in-line sales, as operating leverage exceeded expectations. The company is raising its FY targets, bringing them in line with high consensus expectations. Demand remains robust, evidenced by a meaningful acceleration in order growth to +56% YoY in FY1Q from +16% in FY25, bolstered by +179% growth in medical equipment (likely including one-off large orders) and +111% growth in semiconductor equipment. AI robot orders grew modestly by +25% YoY. We remain optimistic about the favourable revenue opportunities across humanoid robots, space, and semiconductors, as well as the resulting profitability improvements from increased capacity utilisation. We raise our forecasts and increase our target price to ¥9,900 from ¥8,500.

Key highlights from FY1Q results: Beat and raise; strong orders led by semis and medical whilst AI robots also growing

- **FY1Q profits ahead of expectations demonstrating strong operating leverage:** In FY1Q, HDS delivered sales of ¥16.7bn (+24% YoY), operating profit of ¥1.8bn (+1402% YoY) and net profit of ¥1.3bn (net loss of ¥38mn a year ago). Compared with Bloomberg consensus, operating profit/ net profit came in 34%/ 20% ahead of expectations, on inline sales. Refer to [Figure 1](#). Operating leverage was strong in the quarter, with significant OPM improvement to 11% in FY1Q, +10ppt YoY and +3ppt QoQ.
- **FY raised and aligned with high consensus levels:** FY targets have been raised to sales of ¥74.5bn (+25% YoY), operating profit of ¥8.5bn (+40% YoY) and net profit of ¥6bn (+273% YoY). The revised profit targets broadly align with consensus levels, whilst the new sales target is 4% ahead.
- **Strong order growth of +56% YoY led by semis and medical, whilst AI robots also growing:** HDS has disclosed consolidated orders (previously non-consolidate orders only) by end-market for the first time. Consolidated orders rose sharply by +56% in FY1Q, driven mainly by +111% growth from semiconductor equipment and +179% growth from medical equipment. AI

Valuation & Risks

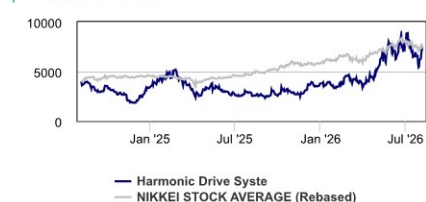
Iris Zheng, CFA
Research Analyst
+852-2203-5884

Key changes

Target price (¥)	8,500 to 9,900	↑	16.5%
------------------	----------------	---	-------

Source: Deutsche Bank

Price/price relative



Performance (%)	1m	3m	12m
Absolute	-19.8	38.4	152.8
NIKKEI STOCK AVERAGE	-5.8	10.4	61.0

Source: Deutsche Bank

Key indicators (FY1)

ROE (%)	7.2
Net debt/equity (%)	-12.6
Book value/share (JPY)	878.42
Price/book (x)	8.1
Net interest cover (x)	129.4
Operating profit margin (%)	11.5

Source: Deutsche Bank

Deutsche Bank AG/Hong Kong

IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. THE CONTENT MAY NOT BE DISTRIBUTED IN THE PEOPLE'S REPUBLIC OF CHINA ("THE PRC") (EXCEPT IN COMPLIANCE WITH THE APPLICABLE LAWS AND REGULATIONS OF PRC), EXCLUDING SPECIAL ADMINISTRATIVE REGIONS OF HONG KONG AND MACAU.



robot orders grew modestly by +25% YoY. Aviation grew +38% YoY, industrial robots +27%, machine tool also +27% and automotive +15%.

- **Japan and North America strong, but China remained weak.** By region, orders grew +126% YoY in North America and +52% in Japan, followed by +37% in Europe. China, however, contracted by -23%, recording four consecutive quarters of declines and implying continued share loss of HDS in the region — China shrunk to only 6% of group sales in FY1Q from ~10% more than a year ago.

Humanoid robot: Long-term prospects remain attractive

AI robot orders grew modestly by +24% YoY to ¥1.25bn (RMB53mn) in FY1Q, after achieving ¥2.7bn (RMB126mn) in FY25, we remain optimistic about the long-term prospects.

HDS is engaging with various humanoid robot customers in the US and Japan. We think the company continues to be a strong contender of strain wave reducers for humanoid robot customers in the US such as **Tesla**, especially considering the **heightened geopolitical sensitivity about humanoids** evident by the recent US ban on foreign-produced mobile robots ([link](#)).

Among reducer suppliers, we believe **HDS holds a clear advantage** due to its industry-leading expertise in reducer manufacturing and its diversified production footprint across Japan, the US, and Germany, which mitigates geopolitical risk. We anticipate that global policymakers will gradually become more aware of supply chain reliance on Chinese suppliers. This could create **opportunities for non-Chinese suppliers**, even if at a higher cost. In our view, HDS holds a commanding position in strain wave reducers for robots globally and represents **a natural non-Chinese alternative**. Apart from strain wave reducers, we note HDS also manufactures planetary reducers.

In addition to reducers, HDS is **developing actuators to enhance value-add**; we estimate that the **average selling price (ASP) of an actuator is approximately 3x that of a reducer**. The company is currently fielding interest from humanoid robot start-ups in Japan and plans to commercialize its actuator products within the next 1.5 years.

Management mentioned before the target to **increase its production capacity in the US** to 10,000-15,000 units per month by the end of 2026, up from the current 8,000-9,000 units.

After posting ¥1.9bn (RMB90mn), representing 3% of group revenue, in FY3/26, **we forecast humanoid robot revenue to more than double to ¥4.2bn (~RMB180mn), or 6% of group revenue, for FY3/27E, and ¥9bn (~RMB380mn), or 10% of group revenue, for FY3/28E**. We consider this forecast undemanding when compared to our humanoid robot revenue estimate for its Chinese reducer peer, **Leaderdrive**, which is RMB100mn for 2025 and RMB350mn for 2026E.

In our SOTP valuation, we continue to **value the humanoid robot component business at RMB18.5bn/¥425bn, or close to ¥4,500 per share, for HDS** ([Figure 3](#) and [Figure 4](#)). Our SOTP valuation assumes 1mn units of humanoid robots globally by 2030E, with HDS capturing a 15% market share of reducers and a 3% share for rotary actuators. This framework is widely adopted by humanoid robot investors, particularly in China, as they seek growth opportunities 5-10 years out, given the



nascent stage of the market. We then discount the value from 2030E back to 2027E.

Revenue share from Aviation & Space to increase; Semis to grow strongly in FY26E and FY27E

HDS has provided a breakdown of its revenue forecast for the next five years by end market during past results ([Figure 5](#)):

- **AI robot** (3% of group sales in FY25, to increase to 13% by FY30E): As mentioned above, we anticipate upside to this forecast.
- **Aviation & Space** (10% of group sales in FY25, to increase to 15% by FY30E): This segment includes gearboxes for aviation, aerospace, and defense applications such as **Mars and moon rovers** (drive systems), **satellites** (solar panel deployment and antenna pointing), **launchers** (fuel and nozzle control systems), **aircraft** (actuators for high-lifting surfaces, position pick-off units, landing gears, and trolley lift), and **helicopters**. HDS products have been adopted by prominent organizations including **NASA**, **Tesat-Spacecom** (for optical communication terminals), **JAXA** (Japan Aerospace and Exploration Agency), and **Airbus**.
- **Semiconductor equipment** (14% of group sales in FY25, to 13% by FY30E): Although HDS does not forecast an increase in the revenue share of semiconductor equipment, we believe **the growth is underestimated**. Our US Semicap Equipment analysts raised their forecast, again, for the global wafer fab equipment (WFE) spend in July, now to US\$145bn (+32% YoY) in 2026E, US\$193bn (+33%) in 2027E and US\$220bn (+14%) in 2028E. We expect risk to these estimates remain to the upside as customers continue to find ways to accelerate capacity buildouts. [Link to DB US Semicap Equipment report.](#)

Raise target price to ¥9,900

We **raise our net profit forecast** by +33%, +31%, 29% for FY3/27, FY3/28, and FY3/29, respectively.

We increase **our target price to ¥9,900 from ¥8,500**. We employ a sum-of-the-parts (**SOTP**) **valuation** methodology, as we believe high-growth areas such as humanoid robots and space should be valued separately from the mature underlying business of traditional industrial robots and machinery. Our target price comprises: **(1) ¥4,866 from the underlying business** (versus ¥3,466 previously), derived by applying approximately 90x one-year forward P/E, which is higher than HDS' long-term average of 78x, as we believe HDS has entered an upcycle; **(2) ¥542 from Space & Aviation** (unchanged), where we use 6.5x FY26E P/S; and **(3) ¥4,492 from humanoid robots** (unchanged), where we look to 2030E and assume a 15% market share for HDS in reducers and 3% for rotary actuators.

Downside risks include: 1) a slow recovery in the traditional industrial robot market globally; 2) slow development of the humanoid robot market, accompanied by intense competition among strain wave reducer manufacturers and the possibility of strain wave reducers being replaced by other transmission devices; and 3) intense price competition, which could affect HDS' margin recovery.



Figure 1: FY1Q operating profit / net profit were ahead of consensus by +34% / +20%, with inline sales

HDS (Yen mn)	FY3/25	1Q FY3/26	2Q FY3/26	1H FY3/26	3Q FY3/26	4Q FY3/26	2H FY3/26	FY3/26	1Q FY3/27		
									Actual	Consensus	Act vs Cons
Speed Reducers	41,735	12,748	10,173	22,921	11,132	13,899	25,031	47,952	16,810		
YoY	22.2%	30.2%	2.9%	16.5%	6.6%	19.6%	13.5%	14.9%	31.9%		
QoQ		9.7%	-20.2%		9.4%	24.9%			20.9%		
Book-to-bill ratio	1.0x	1.2x	0.9x	1.0x	1.0x	1.0x	1.0x	1.0x	1.3x		
Mechatronic products	11,306	2,753	3,845	6,599	3,344	3,719	7,063	13,661	7,313		
YoY	13.8%	33.3%	-5.9%	7.3%	38.0%	36.1%	37.0%	20.8%	165.6%		
QoQ		0.8%	39.7%		-13.0%	11.2%			96.7%		
Book-to-bill ratio	0.8x	1.0x	1.2x	1.1x	0.9x	1.0x	1.0x	1.0x	1.9x		
Total orders	53,041	15,501	14,018	29,520	14,476	17,617	32,093	61,613	24,123		
Orders YoY	20.3%	30.7%	0.4%	14.3%	12.6%	22.7%	17.9%	16.2%	55.6%		
Orders QoQ		8.0%	-9.6%		3.3%	21.7%			36.9%		
Book-to-bill ratio	1.0x	1.1x	1.0x	1.1x	1.0x	1.0x	1.0x	1.0x	1.4x		
Speed Reducers	42,304	10,804	11,101	21,905	10,813	13,616	24,429	46,334	12,790		
YoY	7.3%	11.0%	9.2%	10.0%	1.6%	15.8%	9.1%	9.5%	18.4%		
% of total	76%	80%	77%	79%	75%	78%	77%	78%	77%		
Mechatronic products	13,342	2,692	3,242	5,934	3,532	3,758	7,290	13,224	3,887		
YoY	-18.5%	-17.4%	-4.4%	-10.8%	11.7%	6.6%	9.0%	-0.9%	44.4%		
% of total	24%	20%	23%	21%	25%	22%	23%	22%	23%		
Sales	55,646	13,496	14,343	27,839	14,345	17,374	31,719	59,558	16,681	16,304	2%
Sales YoY	-0.3%	3.8%	5.8%	4.8%	3.9%	13.7%	9.1%	7.0%	23.6%	20.8%	
Gross Profit	14,854	3,829	4,136	7,965	4,508	5,660	10,168	18,133	5,932		
Gross margin	26.7%	28.4%	28.8%	28.6%	31.4%	32.6%	32.1%	30.4%	35.6%		
Gross profit YoY	-5%	12%	25%	18%	16%	34%	25%	22%	55%		
Operating Profit	7	123	343	465	726	1,377	2,102	2,568	1,842	1,374	34%
Operating margin	0.0%	0.9%	2.4%	1.7%	5.1%	7.9%	6.6%	4.3%	11.0%	8.4%	
Operating Profit YoY	-94%	-151%	-186%	-173%	141%	301%	226%	36607%	1402%	1020%	
Recurring Profit	151	132	371	503	755	1,282	2,037	2,540	1,719		
Net Profit	3,474	(38)	355	317	453	839	1,292	1,609	1,275	1,066	20%
Net margin	6.2%	-0.3%	2.5%	1.1%	3.2%	4.8%	4.1%	2.7%	7.6%	6.5%	
Net Profit YoY	-114.0%	-86.5%	-162.7%	-137.3%	-8.7%	-78.1%	-70.1%	-53.7%	-3433.6%	n.a.	
EPS (in Yen)	36.6	(0.4)	3.8	3.4	4.8	8.9	13.6	17.0	13.5		
EPS YoY	n.a.	n.a.	n.a.	-137%	-8%	-78%	-70%	-54%	-67%		
Cash Flow from Operations	7,516			3,881			2,544	6,425			
OCF YoY	-41.0%			22.7%			-41.6%	-14.5%			
OCF as % of net profit	216%			1224%			197%	399%			

Source : Deutsche Bank, company data, Bloomberg Finance LP



Figure 2: Estimate changes, and estimates vs consensus and company guidance

FY (in Yen mn)	FY3/26 Actual	FY3/27E DBE	DBE	Chg %	BBG cons	DBE vs BBG	Co. Guidance	DBE vs Co.	FY3/28E DBE	DBE	Chg %	BBG cons	DBE vs BBG	FY3/29E DBE	DBE	Chg %	BBG cons	DBE vs BBG
Sales	59,558	74,211	69,372	7.0%	71,928	3.2%	74,500	-0.4%	87,260	80,845	7.9%	83,995	3.9%	96,817	89,814	7.6%	93,676	3.4%
Sales YoY	7.0%	24.6%	16.5%		20.8%		25.1%		17.6%	16.5%		16.8%		11.0%	11.1%		11.5%	
o/w growth ex FX	5.3%	22.1%	13.6%						17.6%	16.5%				11.0%	11.1%			
o/w FX	1.7%	2.6%	2.9%						0.0%	0.0%				0.0%	0.0%			
Cost of Goods Sold	(41,424)	(46,600)	(45,025)						(52,135)	(50,194)				(56,053)	(54,220)			
Gross Profit	18,133	27,611	24,347						35,125	30,650				40,764	35,694			
Gross margin	30.4%	37.2%	35.1%						40.3%	37.9%				42.1%	39.6%			
R&D	(3,950)	(4,378)	(4,093)						(5,236)	(4,851)				(5,809)	(5,389)			
Other SG&A	(11,616)	(14,694)	(13,899)						(17,016)	(15,955)				(18,686)	(17,546)			
Total SG&A costs	(15,566)	(19,072)	(17,992)						(22,251)	(20,806)				(24,495)	(22,935)			
Operating Profit	2,568	8,539	6,355	34.4%	8,473	0.8%	8,500	0.6%	12,873	9,845	30.8%	13,556	-5.0%	16,270	12,660	28.5%	16,855	-2.3%
Operating margin	4.3%	11.5%	9.2%		11.8%		11.4%		14.8%	12.2%		16.1%		16.8%	14.1%		17.8%	
Operating leverage - OP	65.5%	40.8%	38.6%		47.7%		39.7%		33.2%	30.4%		42.1%		35.5%	31.4%		32.0%	
Interest & Dividend Income	(44)	(66)	(66)						(79)	(79)				(95)	(95)			
Equity method earnings	(36)	(36)	(36)						(36)	(36)				(36)	(36)			
Other non-operating gain (loss)	153	-	-						-	-				-	-			
Total non-operating Gain (Loss)	(28)	(302)	(102)						(115)	(115)				(131)	(131)			
Recurring Profit	2,540	8,237	6,253	31.7%			8,200	0.6%	12,759	9,730	31.1%			16,139	12,529	28.8%		
Recurring margin	4.3%	11.1%	9.0%				11.0%		14.6%	12.0%				16.7%	13.9%			
Extraordinary Gain (Loss)	(246)	(20)	-						(21)	-				(22)	-			
Pretax Profit	2,294	8,217	6,253	31.4%					12,738	9,730	30.9%			16,117	12,529	28.6%		
Taxes	(685)	(2,301)	(1,813)						(3,821)	(2,919)				(4,835)	(3,759)			
Tax rate	29.9%	28.0%	29.0%						30.0%	30.0%				30.0%	30.0%			
Minority ShareHld Profit or Loss	-	-	-						-	-				-	-			
% of PAT	0.0%	0.0%	0.0%						0.0%	0.0%				0.0%	0.0%			
Net Profit	1,609	5,916	4,440	33.3%	6,127	-3.4%	6,000	-1.4%	8,916	6,811	30.9%	9,807	-9.1%	11,282	8,770	28.6%	11,901	-6.2%
Net margin	2.7%	8.0%	6.4%		8.5%		8.1%		10.2%	8.4%		60.1%		11.7%	9.8%		12.7%	
Net Profit YoY	-53.7%	267.7%	176.0%		280.8%		272.9%		50.7%	53.4%				26.5%	28.8%		74.7%	
EPS (in Yen)	17	62	47	33.3%	66	-5.2%	63	-1.4%	94	72	30.9%	105	-10.1%	119	93	28.6%	127	-6.3%
Shares outstanding ('000)	94,697	94,697	94,697						94,697	94,697				94,697	94,697			
DPS (in Yen)	20	20	20		23				24	18		32		30	23		39	
Payout ratio	117.7%	32.0%	42.7%		34.3%				25.0%	25.0%		30.5%		25.0%	25.0%		30.6%	

Source : Company data, Bloomberg Finance LP, Deutsche Bank estimates

Figure 3: SOTP valuation for HDS

	FY3/27E (Yen mn)				Multiples		Valuation			
	Sales	% of total	Net profit	NPM	P/S	P/E	In Yen mn	Per share (Yen)	% of total	In RMB mn
Scenario: Base case										
Underlying business	62,111	84%	5,111	8.2%	7.4x	90.2x	460,797	4,866	49%	20,035
Space & Aviation	7,900	11%	553	7.0%	6.5x	50,797.0x	51,350	542	5%	2,233
Humanoid robot business	4,200	6%	252	6.0%	101.3x	425,135.2x	425,387	4,492	45%	18,495
HDS group	74,211	100%	5,916	8.0%			937,534	9,900	100%	40,762
Scenario: Bull case										
Underlying business							460,797	4,866	28%	20,035
Space & Aviation							51,350	542	3%	2,233
Humanoid robot business							1,148,614	12,129	69%	49,940
HDS group							1,660,761	17,538	100%	72,207
Scenario: Bear case										
Underlying business							460,797	4,866	79%	20,035
Space & Aviation							51,350	542	9%	2,233
Humanoid robot business							74,056	782	13%	3,220
HDS group							586,203	6,190	100%	25,487

Source : Deutsche Bank estimates



Figure 4: Valuation for the humanoid robot business for HDS

Humanoid robot market forecast for 2030E		Unit	Body reducer	Dexterous hand reducer	Body actuator
Humanoid robot	Unit		1,000,000	1,000,000	1,000,000
Per humanoid robot	Unit		20	22	20
Total unit demand	Unit		20,000,000	22,000,000	20,000,000
Cost per unit	Rmb		1,000	2,000	3,000
Cost per robot	Rmb		20,000	44,000	60,000
Market for humanoid robot	Rmb'mn		20,000	44,000	60,000
Cost per unit	Yen		23,000	46,000	69,000
Cost per robot	Yen		460,000	1,012,000	1,380,000
Market for humanoid robot	Yen'mn		460,000	1,012,000	1,380,000

HDS's humanoid robot business for 2030E		Unit	Scenario		
			Bear	Base	Bull
(1) Body reducer:					
Market share of HDS	%		5%	15%	30%
Strain wave reducer total unit demand for HDS	Unit		1,000,000	3,000,000	6,000,000
Sales for HDS from humanoid robot	Rmb'mn		1,000	3,000	6,000
Net profit for HDS from humanoid robot	Rmb'mn		100	300	600
Net profit margin	%		10%	10%	10%
Sales for HDS from humanoid robot	Yen'mn		23,000	69,000	138,000
Net profit for HDS from humanoid robot	Yen'mn		2,300	6,900	13,800
Valuation					
2030 P/E	Times		40x	40x	40x
WACC	%		7.5%	7.5%	7.5%
Discount factor to 2027	%		80%	80%	80%
Valuation for HDS	Rmb'mn		3,220	9,660	19,319
Valuation for HDS	Yen'mn		74,056	222,169	444,338
(2) Dexterous hand reducer:					
Probability for using reducers for dexterous hands	%		0%	0%	10%
Market share of HDS	%		0%	20%	35%
Reducer total unit demand for HDS	Unit		-	-	770,000
Sales for HDS from humanoid robot	Rmb'mn		-	-	1,540
Net profit for HDS from humanoid robot	Rmb'mn		-	-	231
Net profit margin	%		15%	15%	15%
Valuation					
2030 P/E	Times		40x	40x	40x
WACC	%		7.5%	7.5%	7.5%
Discount factor to 2027	%		80%	80%	80%
Valuation for HDS	Rmb'mn		-	-	7,438
Valuation for HDS	Yen'mn		-	-	171,070
(3) Body actuator:					
Market share of HDS	%		0%	3%	8%
Body actuator total unit demand for HDS	Unit		-	609,800	1,600,000
Sales for HDS from humanoid robot	Rmb'mn		-	1,829	4,800
Net profit for HDS from humanoid robot	Rmb'mn		-	274	720
Net profit margin	%		15%	15%	15%
Valuation					
2030 P/E	Times		40x	40x	40x
WACC	%		7.5%	7.5%	7.5%
Discount factor to 2027	%		80%	80%	80%
Valuation for HDS	Rmb'mn		-	8,836	23,183
Valuation for HDS	Yen'mn		-	203,218	533,206
Humanoid robot total = (1) + (2) + (3)					
Valuation for HDS	Rmb'mn		3,220	18,495	49,940
Valuation for HDS	Yen'mn		74,056	425,387	1,148,614
Valuation for HDS (per share)	Yen		782	4,492	12,129

Source : Deutsche Bank estimates



Figure 5: Long-term targets: HDS guides for the revenue share of AI robot and Space to grow

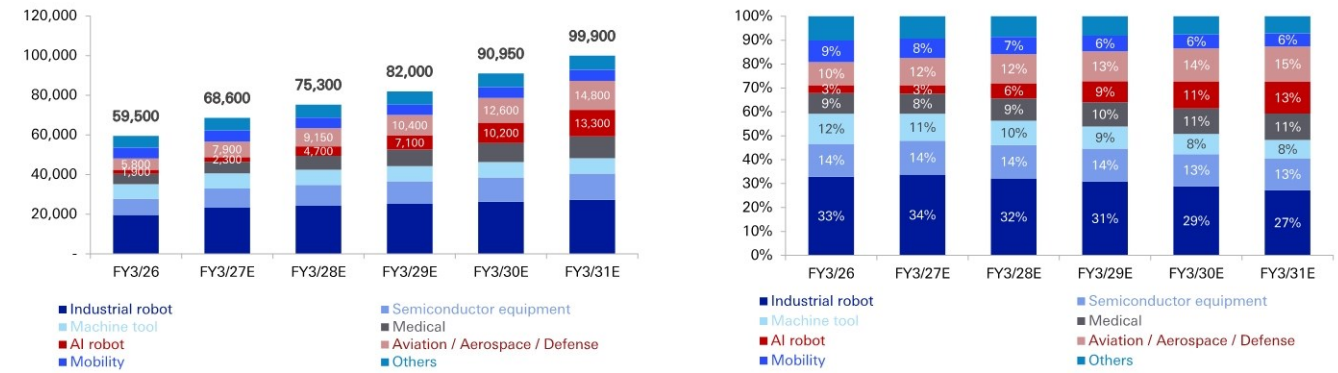
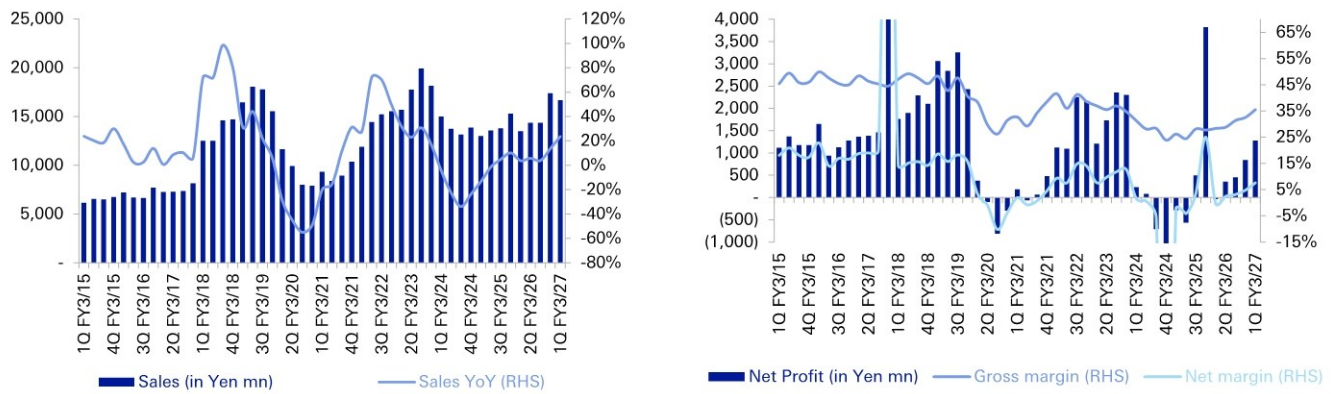


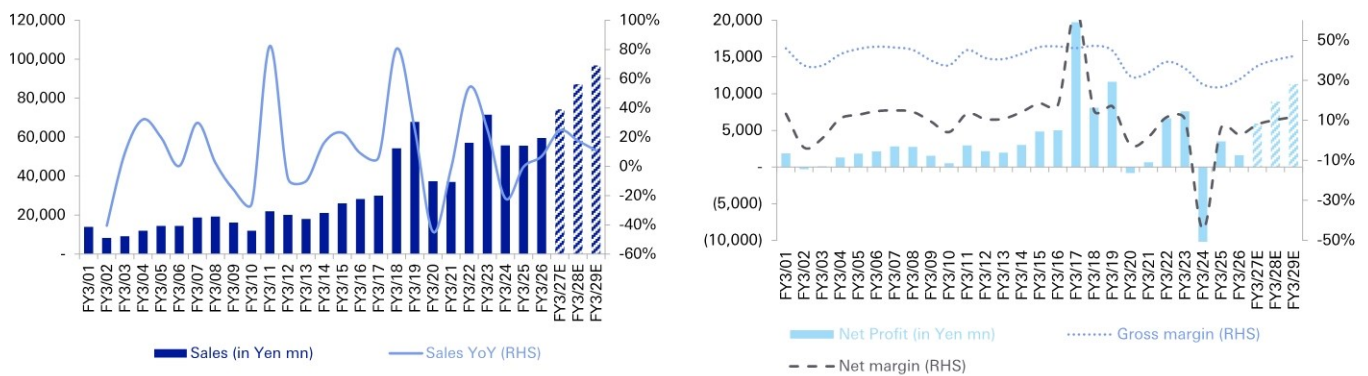


Figure 6: Financial history (quarterly) – HDS has been delivering clear profit recovery in recent quarters



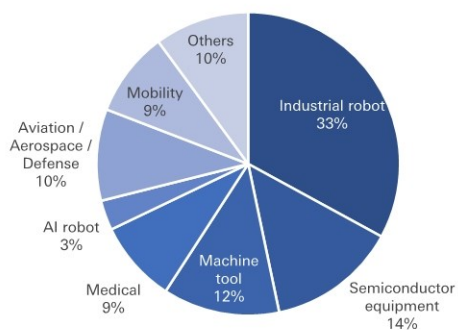
Source : Company data, Deutsche Bank

Figure 7: Financial history (annual) – we forecast revenue and net profit for HDS to improve meaningfully in the coming years



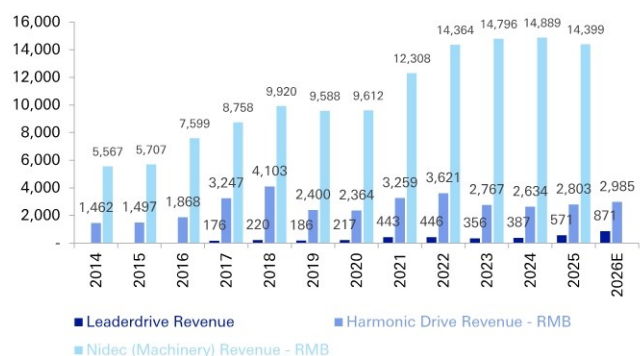
Source : Company data, Deutsche Bank estimates

Figure 8: Split of group consolidated sales of HDS in FY3/26



Source : Company data, Deutsche Bank

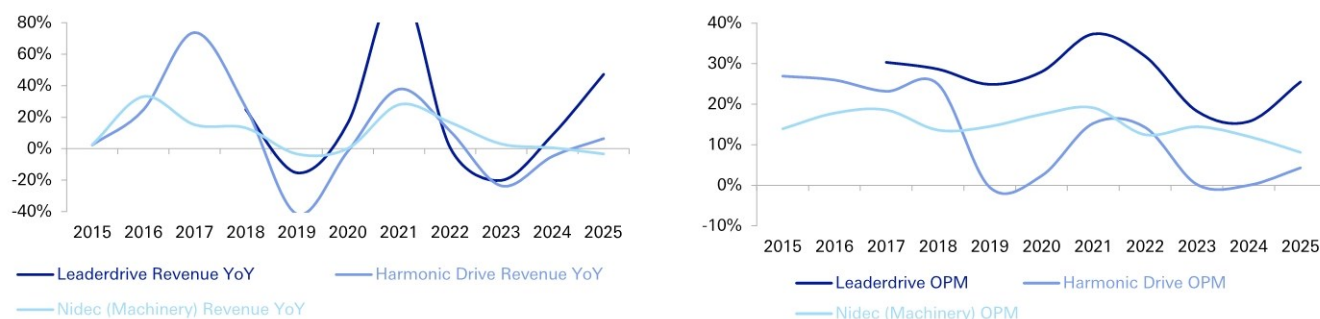
Figure 9: HDS generated 5x revenue from reducers vs Leaderdrive in 2025



Source : Company data for HDS and Leaderdrive, forecasts from Deutsche Bank, Bloomberg Finance LP consensus for Nidec

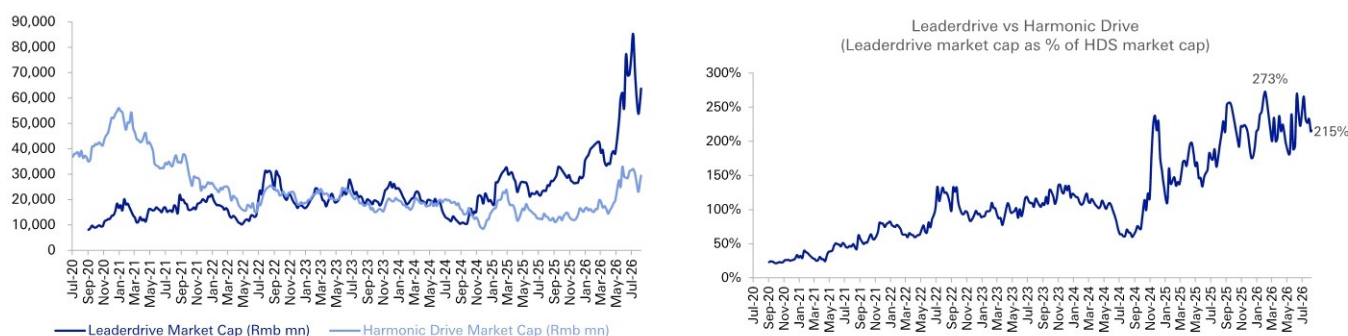


Figure 10: Leaderdrive delivers faster growth; Both Leaderdrive and Nidec generate higher OPM than HDS



Source : Company data for HDS and Leaderdrive, Bloomberg Finance LP consensus for Nidec

Figure 11: Leaderdrive is trading at a premium to Harmonic Drive in terms of market cap



Source : Deutsche Bank, Bloomberg Finance LP

Figure 12: Capacity of HDS: HDS has increased capacity meaningfully during 2018-2022; the current utilisation rate has improved from 50% to 70%, and management targets 80%

Capacity (unins / year)	Mar-18	Mar-22	Sep-22	Mar-25	Mar-26	End of 2026
Industrial machinery						
Hotaka (Japan)	672,000	1,080,000	1,080,000	1,080,000	1,231,200	1,231,200
Ariake (Japan)	0	0	480,000	480,000	547,200	547,200
Germany	120,000	324,000	324,000	353,160	353,160	353,160
US	36,000	96,000	96,000	108,480	108,480	150,000
Total	828,000	1,500,000	1,980,000	2,021,640	2,240,040	2,281,560
Increase		81%	32%	2%	11%	2%
Japan % of total	81%	72%	79%	77%	79%	78%
Germany % of total	14%	22%	16%	17%	16%	15%
US % of total	4%	6%	5%	5%	5%	7%
Automotive products						
Ariake (Japan)	n.a.	720,000	1,080,000	1,080,000	1,080,000	1,080,000
Total	n.a.	720,000	1,080,000	1,080,000	1,080,000	1,080,000
Increase			50%	0%	0%	0%

Source : Company data, Deutsche Bank

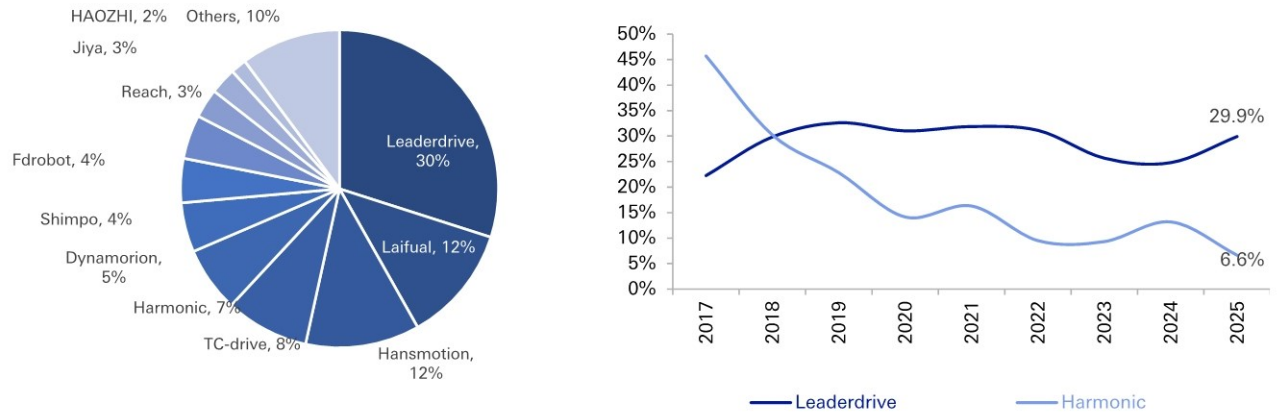


Figure 13: HDS' Chinese peer Leaderdrive is expanding capacity aggressively

Company	Year	Capacity (units / year)	Note
Leaderdrive	2024	590,000	500k units added in late 2023. To add 1mn by 2027
	2027	1,590,000	To add 1m units by 2027
Harmonic Drive	2024	1,980,000	Increased from 808k in 2018 to 1.98mn in 2022
	2025	2,021,640	To increase slightly by March 2025
	2026	2,240,040	To further increase by March 2026

Source : Deutsche Bank, company data

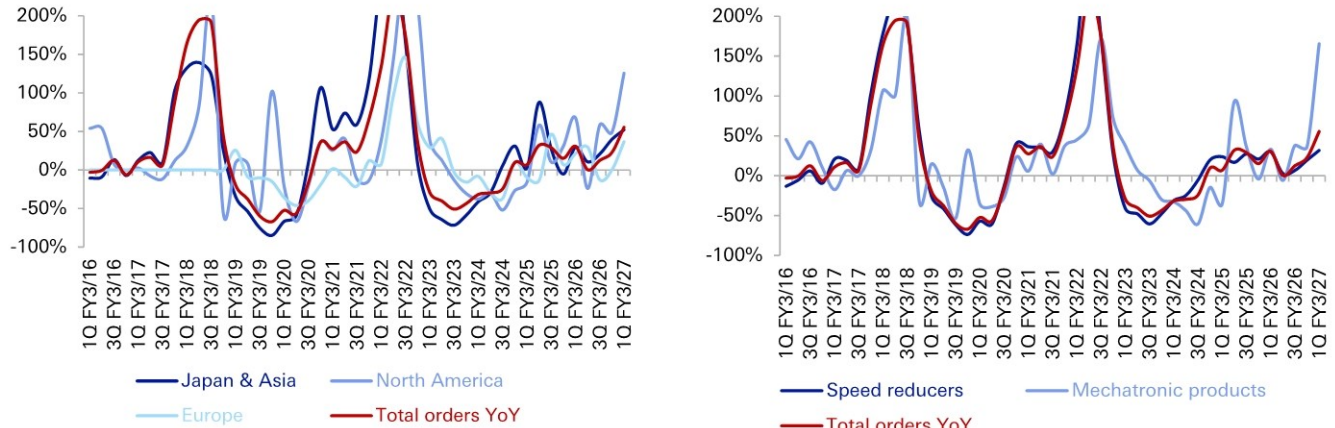
Figure 14: Market share of strain wave reducer in China in 2025



Source : MIR, Deutsche Bank

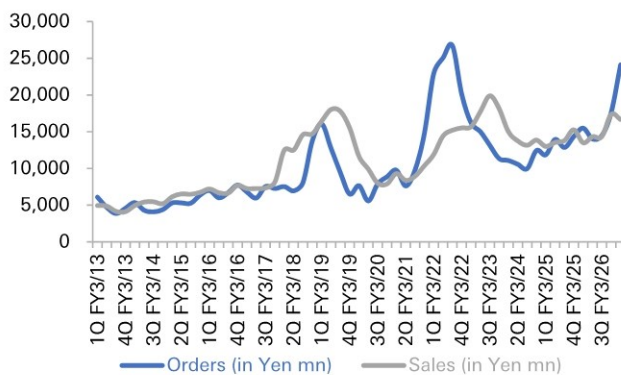


Figure 15: YoY order growth by region (LHS) and by product (RHS) – Total order growth remained robust, led by North America (+126% YoY) and Japan & Asia (+52%)



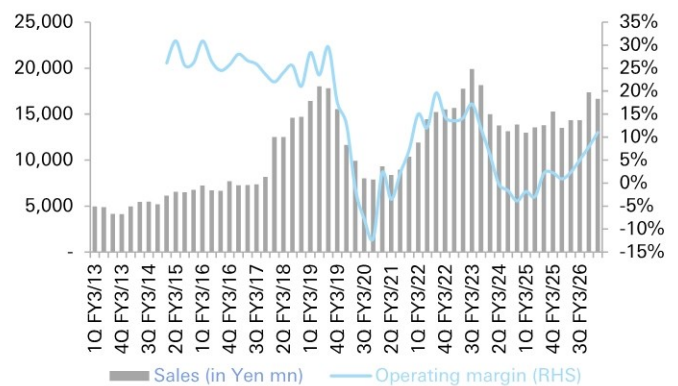
Source : Company data, Deutsche Bank

Figure 16: Orders-to-sales ratio surged to 1.45x in 1Q, the highest since FY3/23



Source : Company data, Deutsche Bank

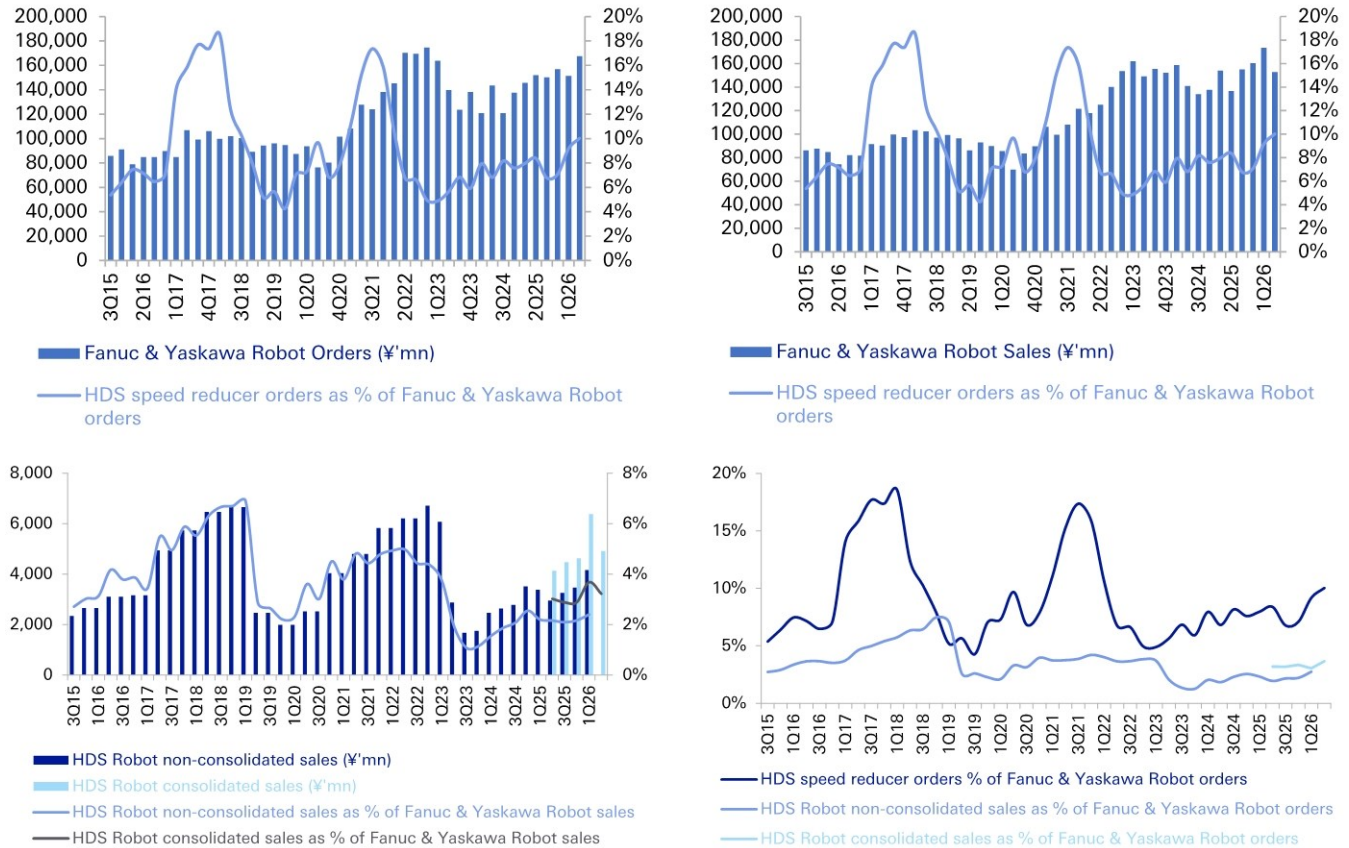
Figure 17: Sales growth accelerated to +24% YoY in 1Q, with OPM improved sequentially to 11%



Source : Company data, Deutsche Bank



Figure 18: HDS's speed reducer orders and robot sales have been at low levels compared with robot orders and sales at FANUC and Yaskawa, but the ratio is improving



Source : Company data, Deutsche Bank



Figure 19: HDS' guidance history: HDS revised up guidance for FY3/27

Group												
Forecast period	Time	Sales ('bn Yen)	vs last forecast	Operating profit ('bn Yen)	vs last forecast	Operating margin	Ordinary profit ('bn Yen)	vs last forecast	Ordinary margin	Profit attributed to owners of parent ('bn Yen)	vs last forecast	NP margin
FY18	4Q FY17	69	-	17.4	-	25.4%	17.6	-	25.7%	11.5	-	16.8%
	1Q FY18	69	-	17.4	-	25.4%	17.6	-	25.7%	11.5	-	16.8%
	2Q FY18	67	(2%)	16.1	(7%)	24.0%	16.5	(6%)	24.6%	10.6	(8%)	15.8%
	3Q FY18	67	-	16.1	-	24.0%	16.5	-	24.6%	10.6	-	15.8%
	Actual result	68	1%	16.9	5%	24.9%	17.5	6%	25.8%	11.6	9%	17.1%
FY19	4Q FY18	42	-	1.0	-	2.4%	1.0	-	2.4%	0.5	-	1.2%
	1Q FY19	42	-	1.0	-	2.4%	1.0	-	2.4%	0.5	-	1.2%
	2Q FY19	37	(13%)	(1.5)	(250%)	-4.1%	(1.3)	(230%)	-3.5%	(1.5)	(400%)	-4.1%
	3Q FY19	37	-	(1.5)	-	-4.1%	(1.3)	-	-3.5%	(1.5)	-	-4.1%
	Actual result	37	2%	(0.2)	(87%)	-0.5%	0.2	(118%)	0.8%	(1.1)	(27%)	-2.9%
FY20	4Q FY19	n.a.	-	n.a.	-	n.a.	n.a.	-	n.a.	n.a.	-	n.a.
	1Q FY20	n.a.	-	n.a.	-	n.a.	n.a.	-	n.a.	n.a.	-	n.a.
	2Q FY20	35	-	(0.8)	-	-2.3%	(0.5)	-	-1.3%	(0.5)	-	-1.3%
	3Q FY20	37	6%	1.0	(225%)	2.7%	1.1	(344%)	3.0%	0.5	(211%)	1.4%
	Actual result	37	0%	0.9	(14%)	2.3%	1.4	24%	3.7%	0.7	32%	1.8%
FY21	4Q FY20	52	-	6.6	-	12.7%	6.7	-	12.9%	4.7	-	9.0%
	1Q FY21	58	12%	9.5	44%	16.4%	9.5	42%	16.4%	6.6	40%	11.4%
	2Q FY21	57	(3%)	8.2	(14%)	14.5%	8.2	(14%)	14.5%	5.5	(17%)	9.7%
	3Q FY21	57	-	8.2	-	14.5%	8.2	-	14.5%	5.5	-	9.7%
	Actual result	57	1%	8.7	7%	15.3%	9.1	11%	16.0%	6.6	21%	11.6%
FY22	4Q FY21	75	-	13.0	-	17.3%	13.0	-	17.3%	9.0	-	12.0%
	1Q FY22	75	-	13.0	-	17.3%	13.0	-	17.3%	9.0	-	12.0%
	2Q FY22	74	(1%)	11.0	(15%)	14.9%	11.3	(13%)	15.3%	7.5	(17%)	10.1%
	3Q FY22	71	(4%)	9.8	(11%)	13.8%	10.1	(11%)	14.2%	6.7	(11%)	9.4%
	Actual result	72	1%	10.2	4%	14.3%	10.8	7%	15.0%	7.6	13%	10.6%
FY23	4Q FY22	n.a.	-	n.a.	-	-	n.a.	-	-	n.a.	-	-
	1Q FY23	n.a.	-	n.a.	-	-	n.a.	-	-	n.a.	-	-
	2Q FY23	55	-	(0.4)	-	-0.7%	(0.2)	-	-0.4%	(0.8)	-	-1.5%
	3Q FY23	55	-	(0.4)	-	-0.7%	(0.2)	-	-0.4%	(0.8)	-	-1.5%
	Actual result	56	1%	0.1	(131%)	0.2%	0.6	(385%)	1.0%	(24.8)	3001%	-44.5%
FY24	4Q FY23	59	-	2.7	-	4.6%	2.7	-	4.6%	2.0	-	3.4%
	1Q FY24	59	-	2.7	-	4.6%	2.7	-	4.6%	2.0	-	3.4%
	2Q FY24	54	(8%)	(0.4)	(115%)	-0.7%	(0.5)	(119%)	-0.9%	(0.4)	(120%)	-0.7%
	3Q FY24	54	-	(0.4)	-	-0.7%	(0.5)	-	-0.9%	3.6	(1000%)	6.7%
	Apr-25	56	3%	0.0	(110%)	0.1%	0.1	(128%)	0.3%	3.5	(4%)	6.2%
	Actual result	56	0%	0.0	(85%)	0.0%	0.2	8%	0.3%	3.5	1%	6.2%
FY25	4Q FY24	n.a.	-	n.a.	-	n.a.	n.a.	-	n.a.	n.a.	-	n.a.
	1Q FY25	57	-	1.5	-	2.6%	1.5	-	2.6%	1.3	-	2.3%
	2Q FY25	57	-	1.5	-	2.6%	1.5	-	2.6%	1.3	-	2.3%
	3Q FY25	57	-	1.5	-	2.6%	1.5	-	2.6%	1.3	-	2.3%
	Apr-26	60	4%	2.5	67%	4.2%	2.5	67%	4.2%	1.6	22%	2.7%
	Actual result	60	0%	2.6	3%	4.3%	2.5	2%	4.3%	1.6	2%	2.7%
FY26	4Q FY25	68	-	6.2	-	9.1%	6.2	-	9.1%	4.5	-	6.6%
	1Q FY26	75	10%	8.5	37%	11.4%	8.2	32%	11.0%	6.0	33%	8.1%

Source : Company data, Deutsche Bank

Figure 20: HDS has a history of missing its medium-term targets; the company has lowered its FY30E OPM target

Yen/bn	FY12-14			FY15-17			FY18-20			FY21-23			FY26			2030	
	Target	FY14 actual	Act vs Target	Target	FY17 actual	Target	Target	FY20 actual	Target	Target	FY23 actual	Target	Target	FY26 actual	Target	Target	Old tgt
Net sales	26	25.9	0%	35	54.3	55%	100	37	-63%	70	55.7	-20%	90	59.6	-34%	100+	100+
Operating income	7.5	7	-7%	9.5	12.5	32%	26	0.8	-97%	15	0.13	-99%	13.5-18	2.57	-84%	>15	15-20
Margin	28.8%	27.0%	-	27.1%	23.0%	-	26.0%	2.2%	-	21.4%	0.2%	-	15-20%	4.3%	-	>15%	15-20%
ROE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	>10%	>10%
Dividend payout ratio	-	27.2%	-	-	31.5%	-	290.6%	-	30.0%	-7.7%	-	-	30%+	117.7%	-	>35%	>30%
Capex (3 years total)	-	5.1	-	10	17.6	76%	71	33.7	-53%	15	19.8	32%	27.5	15.6	-43%	-	-
D&A (3 years total)	-	3.5	-	6	9.1	52%	25.5	21.6	-15%	9	25	178%	-	-	-	-	-
R&D (3 years total)	-	3.6	-	4.8	4.9	2%	9.4	7.1	-24%	-	9.9	-	12.3	11.3	-8%	-	-

Source : Company data, Deutsche Bank



Figure 21: Share buyback program

		Shares repurchased	Amount repurchased (bn yen)	Average repurchase price	shares as % of outstanding shares
Announcement	Jan-25 to Jun-25	400,000	1.2	3,000	0.4%
	Feb-25	37,200	0.2	4,839	0.0%
	Mar-25	52,400	0.2	4,016	0.1%
Tranche	Apr-25	104,200	0.3	2,837	0.1%
	May-25	83,200	0.3	3,408	0.1%
	Jun-25	77,400	0.2	2,976	0.1%
Result	Jun-25	354,400	1.2	3,386	0.4%
Announcement	Dec-23 to Jan-24	700,000	2.0	2,857	0.7%
	Dec-23	24,700	0.1	3,783	0.0%
Tranche	Jan-24	80,100	0.3	3,793	0.1%
Result	Jan-24	104,800	0.4	3,791	0.1%
Announcement	May-22 to Sep-22	2,000,000	5.0	2,500	2.1%
	Jun-22	588,700	2.0	3,379	0.6%
Tranche	Sep-22	626,700	3.0	4,804	0.7%
Result	Sep-22	1,215,400	5.0	4,114	1.3%

Source : Company data, Deutsche Bank

Figure 22: HDS's one-year forward P/E



Source : Bloomberg Finance LP, Deutsche Bank

Link to previous reports:

- **Japan machine tool order update:** [June 2026 \(detailed\) - 30 Jul 2026](#)
- **Humanoid Robot:** [US bans foreign-made robotics \(humanoid, quadruped, mobile robots\) - 29 Jul 2026](#)
- **Rise of Humanoid Robots (IV):** [Takeaways from a Chinese humanoid robot expert - 27 Jul 2026](#)
- **Humanoid Robot Pulse:** [US activities stepping up; China scaling fast; Supportive policies \(2Q edition\) - 16 Jul 2026](#)
- **Harmonic Drive Systems:** [Guidance raise on higher sales and improved utilisation; Recovery on track; Buy - 24 Apr 2026](#)
- **China Automation Market Monthly tracker:** [June stayed solid, finishing 2Q26 stronger than 1Q26; Growth could decelerate in 2H26E - 14 Jul 2026](#)
- **Catalyst road map:** [for next 3M - 8 Jul 2026](#)
- **Humanoid Robot:** [Comparing Unitree, UBTECH, DEEP, Dobot and Leju:](#)



[Opportunities and risks coexist - 15 Jun 2026](#)

- **Expert call:** [takeaways on SMC and global automation and semis demand - 13 Mar 2026](#)
- **Humanoid Robot (III):** [Six visions for 2026 – Scaling, iterating and diversifying - 6 Mar 2026](#)
- **Harmonic Drive Systems:** [Upgrade to BUY; Humanoid opportunity under-appreciated; Recovery on path - 11 Feb 2026](#)
- **Rise of Humanoid Robots:** [Takeaways from expert call with Humanoid - 4 Dec 2025](#)
- **2026 outlook:** [More positives than negatives - 25 Nov 2025](#)
- **Harmonic Drive Systems:** [Humanoid orders improved; US factory acquired for expansion; Hold - 19 Nov 2025](#)
- **Annual benchmarking of global peers:** [Some hopes and some gaps - 17 Sep 2025](#)
- **Harmonic Drive Systems:** [FY1Q profit and FY profit guidance miss; Humanoid design update is key - 7 Aug 2025](#)
- **Harmonic Drive Systems:** [Two surprising and puzzling gaps; FY4Q results mixed; Reiterate Hold - 26 May 2025](#)
- **Humanoid Robot (II):** [Go big and go home - 14 Feb 2025](#)
- **Harmonic Drive Systems:** [Updating humanoid robot forecast and 3Q result; Reiterate Hold – 14 Feb 2025](#)
- **APAC Automation & Machinery Initiation:** [Stay selective; Humanoid robots lift growth prospects - 23 July 2024](#)
- **APAC Industrial Automation Initiation:** [Identifying winners amid rising competition from China - 19 Feb 2024](#)



Model updated: 07 August 2026

Running the numbers

Japan

Japan

Machinery

Harmonic Drive Systems

Reuters: 6324.T

Bloomberg: 6324 JP

Buy

Price (6 Aug 26) JPY 7,130.0

Target Price JPY 9,900.0

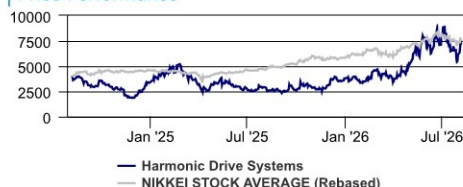
52 Week range JPY 2,328 - 8,890

Market cap (m) JPYbn 677.2
USDm 4,274.9

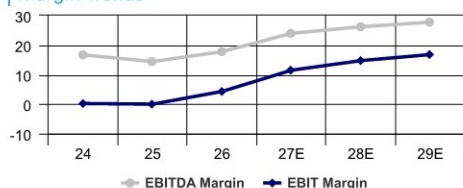
Company Profile

Harmonic Drive Systems (HDS) manufactures motion control products including speed reducers (HarmonicDrive[®], AccuDrive[®], and HarmonicPlanetary[®]) and mechatronic products that combine speed reducers, motors, sensors, drivers, controllers and other system elements to provide the motion desired by customers.

Price Performance



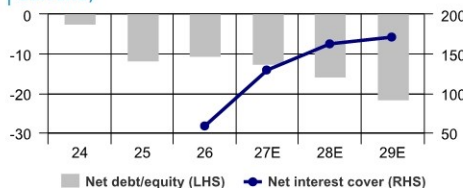
Margin Trends



Growth & Profitability



Solvency



Iris Zheng, CFA

+852 2203 5884

iris.zheng@db.com

Fiscal year end 31-Mar

Financial Summary

	2024	2025	2026	2027E	2028E	2029E
DB EPS (JPY)	-261.00	36.57	16.99	62.48	94.16	119.14
Reported EPS (JPY)	-261.00	36.57	16.99	62.48	94.16	119.14
DPS (JPY)	20.00	20.00	20.00	20.00	23.54	29.78
BVPS (JPY)	835.4	831.1	848.9	878.4	936.6	1,018.3
Weighted average shares (m)	95	95	95	95	95	95
Average market cap (JPYbn)	379	351	302	677	677	677
Enterprise value (JPYbn)	377	342	294	667	663	657

Valuation Metrics

P/E (DB) (x)	nm	101.0	187.7	114.1	75.7	59.8
P/E (Reported) (x)	nm	101.0	187.7	114.1	75.7	59.8
P/BV (x)	4.77	3.80	4.08	8.12	7.61	7.00
FCF Yield (%)	2.1	0.8	0.5	0.8	1.0	1.5
Dividend Yield (%)	0.5	0.5	0.6	0.3	0.3	0.4
EV/Sales (x)	6.8	6.1	4.9	9.0	7.6	6.8
EV/EBITDA (x)	40.5	42.5	27.7	37.5	29.0	24.4
EV/EBIT (x)	nm	nm	114.4	78.1	51.5	40.4

Income Statement (JPYbn)

Sales revenue	55.8	55.6	59.6	74.2	87.3	96.8
Gross profit	15.6	14.9	18.1	27.6	35.1	40.8
EBITDA	9.3	8.0	10.6	17.8	22.9	26.9
Depreciation	9.2	8.0	8.0	9.3	10.0	10.6
Amortisation	0.0	0.0	0.0	0.0	0.0	0.0
EBIT	0.1	0.0	2.6	8.5	12.9	16.3
Net interest income(expense)	0.2	0.3	0.0	-0.1	-0.1	-0.1
Associates/affiliates	0.0	0.0	0.0	0.0	0.0	0.0
Exceptionals/extraordinary	-28.2	4.6	-0.2	0.0	0.0	0.0
Other pre-tax income/(expense)	0.3	-0.1	0.0	-0.2	0.0	0.0
Profit before tax	-27.6	4.8	2.3	8.2	12.7	16.1
Income tax expense	-2.8	1.3	0.7	2.3	3.8	4.8
Minorities	0.0	0.0	0.0	0.0	0.0	0.0
Other post-tax income/(expense)	0.0	0.0	0.0	0.0	0.0	0.0
Net profit	-24.8	3.5	1.6	5.9	8.9	11.3
DB adjustments (including dilution)	0.0	0.0	0.0	0.0	0.0	0.0
DB Net profit	-24.8	3.5	1.6	5.9	8.9	11.3

Cash Flow (JPYbn)

Cash flow from operations	12.7	7.5	6.4	11.2	15.6	20.0
Net Capex	-4.9	-4.9	-4.9	-5.9	-8.7	-9.7
Free cash flow	7.8	2.6	1.5	5.3	6.8	10.3
Equity raised/(bought back)	-0.4	-0.4	-0.8	-0.5	-0.6	-0.6
Dividends paid	-2.6	-1.9	-1.9	-2.6	-2.8	-2.9
Net inc/(dec) in borrowings	-4.5	-2.8	-2.4	0.0	0.0	0.0
Other investing/financing cash flows	-1.3	6.5	-0.3	0.0	0.0	0.0
Net cash flow	-1.0	4.0	-3.8	2.1	3.4	6.8
Change in working capital	5.1	-1.1	-2.9	-4.0	-3.4	-1.9

Balance Sheet (JPYbn)

Cash and other liquid assets	20.3	24.9	21.6	23.7	27.1	33.9
Tangible fixed assets	48.1	46.0	45.3	41.9	40.7	39.7
Goodwill/intangible assets	12.9	12.2	12.6	12.6	12.6	12.6
Associates/investments	0.0	0.0	0.0	0.0	0.0	0.0
Other assets	37.8	30.6	32.0	36.2	39.9	42.0
Total assets	119.1	113.6	111.4	114.5	120.3	128.3
Interest bearing debt	18.4	15.6	13.2	13.2	13.2	13.2
Other liabilities	21.3	19.1	17.8	18.1	18.4	18.6
Total liabilities	39.7	34.7	31.0	31.3	31.6	31.9
Shareholders' equity	79.4	78.9	80.4	83.2	88.7	96.4
Minorities	0.0	0.0	0.0	0.0	0.0	0.0
Total shareholders' equity	79.4	78.9	80.4	83.2	88.7	96.4
Net debt	-1.9	-9.3	-8.3	-10.5	-13.9	-20.7

Key Company Metrics

Sales growth (%)	-22.0	-0.3	7.0	24.6	17.6	11.0
DB EPS growth (%)	na	na	-53.5	267.7	50.7	26.5
EBITDA Margin (%)	16.7	14.4	17.8	24.0	26.2	27.8
EBIT Margin (%)	0.2	0.0	4.3	11.5	14.8	16.8
Payout ratio (%)	nm	54.7	117.7	32.0	25.0	25.0
ROE (%)	-27.1	4.4	2.0	7.2	10.4	12.2
Capex/sales (%)	8.8	8.8	8.2	8.0	10.0	10.0
Capex/depreciation (x)	0.5	0.6	0.6	0.6	0.9	0.9
Net debt/equity (%)	-2.4	-11.8	-10.4	-12.6	-15.7	-21.5
Net interest cover (x)	nm	nm	58.4	129.4	162.6	171.2

Source: Company data, Deutsche Bank estimates



Appendix 1

Important Disclosures

*Other information available upon request

Disclosure checklist			
Company	Ticker	Recent price*	Disclosure
Harmonic Drive Systems	6324.T	7130.0 (JPY) 06 Aug 2026	NA

*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

Important Disclosures Required by U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States. See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

Important Disclosures Required by Non-U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States. See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

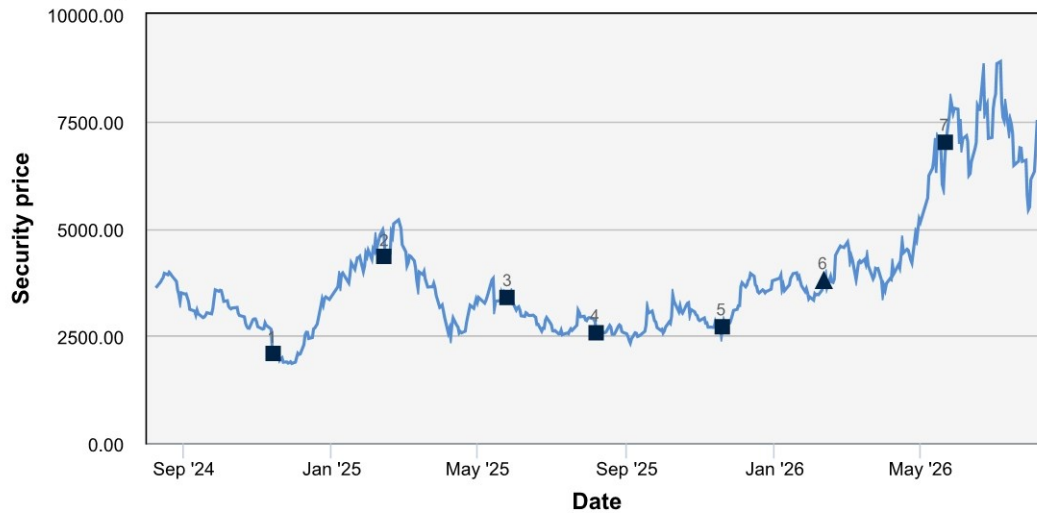
Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s) about the subject issuer and the securities of the issuer. In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Iris Zheng.



Historical recommendations and target price: Harmonic Drive Systems (6324.T)

(as of 08/06/2026)



Current Recommendations

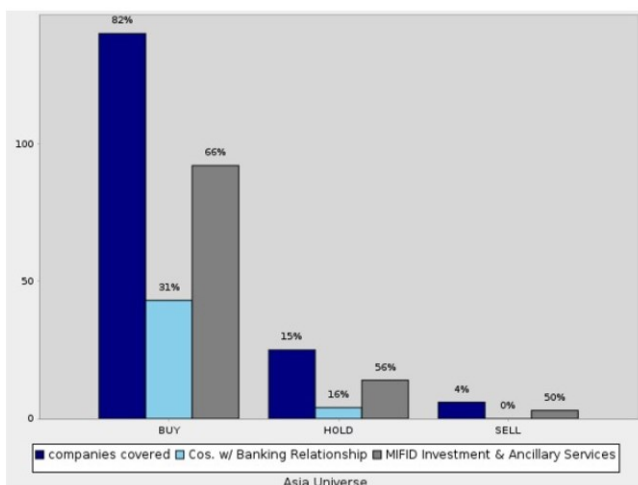
Buy
Hold
Sell
Not Rated
Suspended Rating

** Analyst is no longer at Deutsche Bank

1.	11/14/2024	Hold, Target Price Change JPY 2629.00, Current Price JPY 2087.00 Iris Zheng, CFA	5.	11/19/2025	Hold, Target Price Change JPY 3005.00, Current Price JPY 2708.00 Iris Zheng, CFA
2.	02/14/2025	Hold, Target Price Change JPY 4835.00, Current Price JPY 4350.00 Iris Zheng, CFA	6.	02/11/2026	Upgraded to Buy, Target Price Change JPY 4840.00, Current Price JPY 3800.00 Iris Zheng, CFA
3.	05/26/2025	Hold, Target Price Change JPY 4050.00, Current Price JPY 3390.00 Iris Zheng, CFA	7.	05/22/2026	Buy, Target Price Change JPY 8500.00, Current Price JPY 7010.00 Iris Zheng, CFA
4.	08/07/2025	Hold, Target Price Change JPY 3000.00, Current Price JPY 2565.00 Iris Zheng, CFA			



Equity rating dispersion and banking relationships



Equity Rating and Dispersion Key

The Equity Rating Dispersion Chart depicts the following:

The proportion of recommendations that are rated "buy", "sell" and "hold" over the previous 12 months. This is shown for securities issued in the stated region e.g. "Europe Universe". See rating definitions below. This is represented by the "Companies Covered" bars in the chart. The percentage value displayed above the bar is the proportion as a percentage. E.g. 50% above the "buy" / "Companies Covered" bar means that 50% of DB's equity research covered companies over the past 12 months have a "buy" rating.

Next to each of the three respective bars showing the proportion of "buy", "sell" and "hold" recommendations we provide two additional bars to show:

- The proportion of "buy", "sell" or "hold" recommendations where Deutsche Bank and/or Affiliates provided MIFID Investment or Ancillary Services in the past 12 months. This is represented in the "MIFID Investment and Ancillary Services" bar. The percentage value displayed above the bar shows the proportion of Companies Covered with the given rating where DB has also provided MIFID Investment and Ancillary Services in the past 12 months. E.g. 50% above the "Cos. w/ MIFID Investment and Ancillary Services" bar means 50% of the Companies Covered with the rating stated have also received MIFID Investment and Ancillary Services from DB.

- The proportion of "buy" (or "sell" or "hold") recommendations where Deutsche Bank and/or Affiliates has provided Investment Banking services in the past 12 months for which it has received compensation. The percentage value displayed above the bar shows the proportion of Companies Covered with the stated rating where DB has also provided Investment Banking services in the past 12 months. E.g. 50% above the "Cos. w/ Investment Banking relationship" bar means 50% of the Companies Covered with the rating stated also have an Investment Banking Relationship with DB.

Buy: Based on a current 12-month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

Newly issued research recommendations and target prices supersede previously published research.



Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (<https://research.db.com/Research/>), and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.

The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (<https://research.db.com/Research/>) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.



Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options', at <https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document>. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at <https://research.db.com/Research/> on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report, and the entities for which they act and/or are affiliated with, may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.



India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: <https://www.db.com/ir/en/annual-reports.htm>. For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures



and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD000000000521304.xhtml . Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG. It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2026 Deutsche Bank AG



David Folkerts-Landau

Group Chief Economist and Global Head of Research

Pam Finelli
COO and Head of Fixed
Income Research

Steve Pollard
Global Head of Company
Research and Sales

Jim Reid
Global Head of Macro and
Thematic Research

Tim Rokossa
Head of European
Company Research

Matthew Barnard
Head of Americas
Company Research

Debbie Jones
Global Head of
Sustainability and Data
Innovation, Research

Robin Winkler
Head of German Macro
Research

Sameer Goel
Global Head of EM &
APAC Research

Francis Yared
Global Head of Rates
Research

George Saravelos
Global Head of FX
Research

Peter Hooper
Vice-Chair of Research

Nilendra de-Mel
Head of APAC & Middle
East Product
Development

International Production Locations

Deutsche Bank AG
Deutsche Bank Place
Level 16
Corner of Hunter & Phillip
Streets
Sydney, NSW 2000
Australia
Tel: (61) 2 8258 1234

Deutsche Bank AG
Equity Research
Mainzer Landstrasse 11-17
60329 Frankfurt am Main
Germany
Tel: (49) 69 910 00

Deutsche Bank AG
Filiale Hongkong
International Commerce
Centre,
1 Austin Road West, Kowloon,
Hong Kong
Tel: (852) 2203 8888

Deutsche Securities Inc.
1-3-1 Azabudai
Azabudai Hills Mori JP Tower
Minato-ku, Tokyo 106-0041
Japan
Tel: (81) 3 6730 1000

Deutsche Bank AG
21 Moorfields
London EC2Y 9DB
United Kingdom
Tel: (44) 20 7545 8000

Deutsche Bank Securities Inc.
The Deutsche Bank Center
1 Columbus Circle
New York, NY 10019
Tel: (1) 212 250 2500

Deutsche Bank AG
Filiale Singapur
One Raffles Quay, South
Tower,
Singapore 048583
Tel: +65 6423 8001
